Differential Expression Analysis of TCGA Breast Cancer Data

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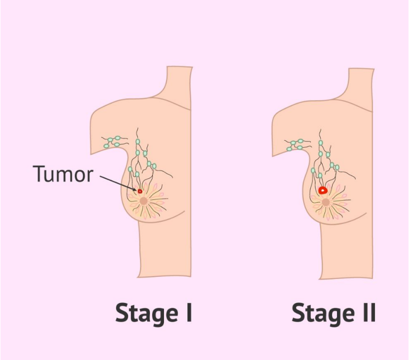
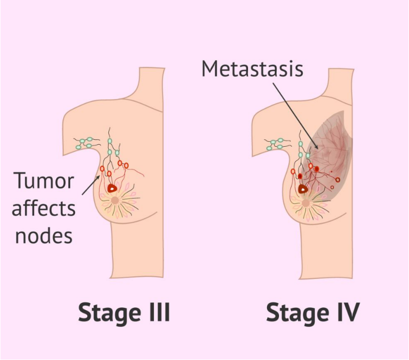
Background

- Breast cancer is one of the most prevalent and biologically complex diseases globally, affecting 1 in 8 women (***ACS, 2026***).
- It is **highly heterogeneous**, which makes diagnosis and targeted treatment a significant clinical challenge.
- Identifying **differentially expressed genes (DEGs)** can reveal candidate biomarkers for more accurate diagnosis and treatment.





Goal:

Determine which genes are most significantly dysregulated between primary tumor and healthy adjacent tissue in the TCGA-BRCA cohort.

Data

- **Source:** The Cancer Genome Atlas (TCGA), Breast Invasive Carcinoma (BRCA) cohort
  - From the Genomic Data Commons (GDC) Portal
- **Package:** TCGAbiolinks (Bioconductor)
  - Used to query and download directly into R
- **Data type**
  - **RNA-sequencing raw counts** (reads mapped to the genome via STAR alignment tool)
  - **Clinical metadata** (PAM50 subtype, staging, vital status)
- **Size:** 1,111 primary tumor and 113 solid tissue normal samples
  - 60,660 genes across 1,224 samples

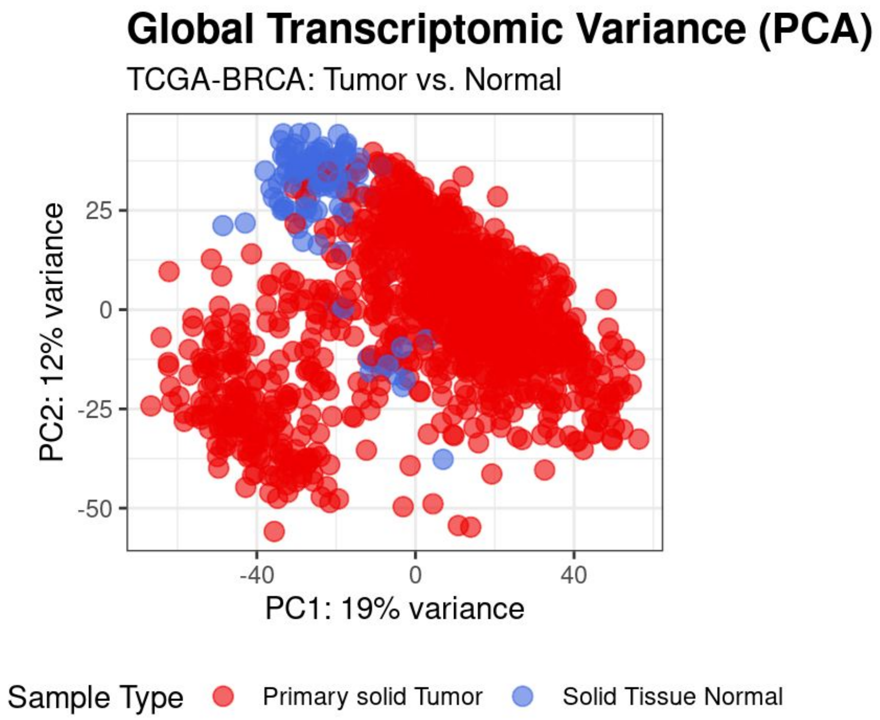
Methods

- **Data wrangling/filtering:** tidyverse (dplyr, tidyr)
- **Pre-filtering:** Removed low-count genes (rowSum < 10) to reduce noise and improve power.
- **Modeling:** DESeq2 (differential expression modeling)
  - Handles overdispersion in count data via negative binomial model, normalizes across samples automatically
- **Thresholds for DEGs:** adjusted p-value < 0.01 and $|\log_2 \text{fold change}| > 2$
- **Transformation:** Variance Stabilizing Transformation (VST) for downstream visualizations
  - Volcano Plot (EnhancedVolcano), Heatmap (ComplexHeatmap + viridis), Boxplots and PCA (ggplot2)
- **ID mapping:** converted Ensembl IDs to gene symbols for interpretability (org.Hs.eg.db)

Results

Figure 1: PCA Plot

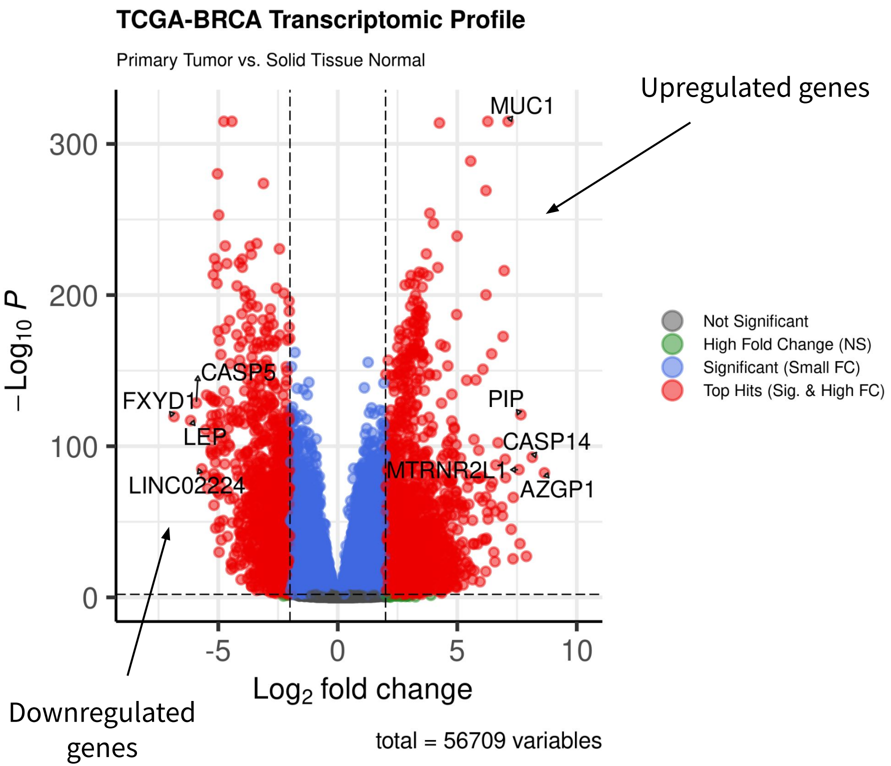
- Shows that the data is high quality and biologically consistent.
- PC1 and PC2 together explain nearly a third of total variance.
- Normal samples (blue) cluster tightly
- Tumor samples (red) show wide spread → reflects tumor heterogeneity
- Few blue dots “drifted” toward the red cluster.
    - Means that a small # of normal samples showed tumor-like expression profiles.



Results

Figure 2: Volcano Plot

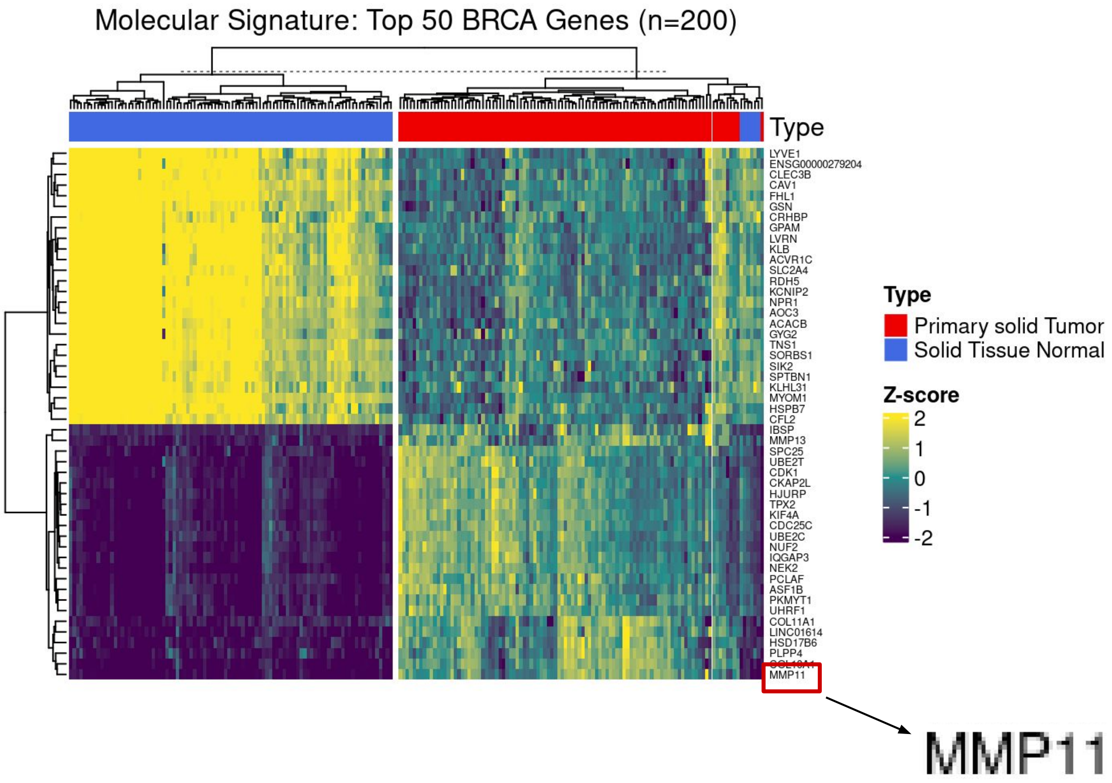
- Visualizes the differential expression of over 56,000 genes.
- Top hits were defined by p-adj < 0.01 and $|\log_2 FC| > 2$
- **Most significantly upregulated genes**
  - **MUC1:** a known oncogene
  - **PIP:** tissue marker, associated with tumor proliferation
- **Most significantly downregulated genes**
  - **FXYD, LEP, CASP5:** Genes associated with normal tissue function



Results

Figure 3: Top 50 DEGs Heatmap

- Shows the top 50 differentially expressed genes across a random subsample of 200 samples (100 for tumor, 100 for normal) for visual clarity.
- Hierarchical clustering cleanly separates tumor from normal with two distinct gene expression blocks.
- Bright yellow = high expression (upregulation)
- Deep purple = low expression (downregulation)

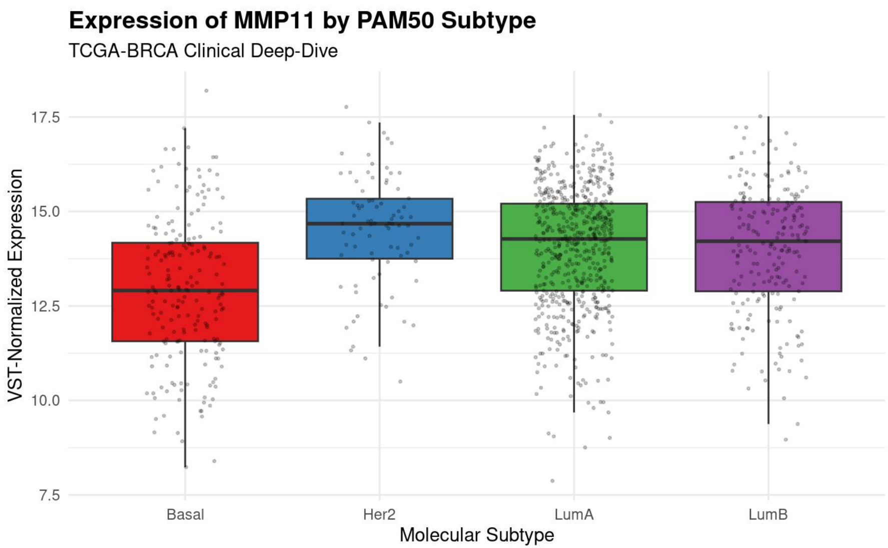


Results

MMP11: Matrix Metalloproteinase, an enzyme that dissolves extracellular matrix (‘glue’ that holds cells in place)

Figure 4: MMP11 Boxplots

- MMP11 expression stratified by PAM50 molecular subtype (Basal, Her2, LumA, LumB)
- Highest median expression in Her-2 subtype.
- Basal subtype shows notably lower MMP11 expression vs. Her2/LumA/LumB, suggesting subtype-specific regulation.
- Broad distribution across all groups suggests that matrix remodeling is a fundamental feature of the malignant phenotype in breast cancer.



Conclusion

- Successfully identified thousands of statistically robust DEGs between tumor and normal tissue.
- Tumor vs. Normal status was the primary driver of gene expression differences, regardless of individual patient backgrounds.
- Identified MMP11 as a candidate biomarker for breast tumor vs. normal tissue.
    - Expression varied across PAM50 subtypes, suggesting that it’s subtype dependent.
- **Limitation**
    - Imbalanced cohort (1,111 tumor vs. 113 normal)
- **Future Directions**
    - Perform Kaplan-Meier survival analysis to see if patients with the highest MMP11 expression have lower overall survival rates.

References

American Cancer Society. (2024). *How common is breast cancer?* https://www.cancer.org/cancer/types/breast-cancer/about/how-common-is-breast-cancer.html

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